

Patuakhali Science and Technology University

Faculty of Computer Science and Engineering

CCE 320 : Software Development Project-II

Project Title: CareerLens

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CareerLens

AI Resume Analyzer & Career Preparation System—Bangladesh Focused



0.1 Abstract

CareerLens AI is an AI-powered platform designed to assist job seekers in Bangladesh in analyzing resumes, identifying skill gaps, and preparing for high-demand job roles. The system uses NLP (TF-IDF, cosine similarity) to evaluate resumes against job descriptions and generate ATS compatibility scores. It also provides Bangladesh-specific job market insights, skill preparation roadmaps, and a progress tracker to measure job readiness.

0.2 Objectives

- Analyze resumes using AI and NLP to calculate ATS compatibility.
- Identify missing skills relevant to target jobs.
- Provide Bangladesh-focused job market insights.
- Generate structured skill preparation roadmaps for trending roles.
- Track user progress and calculate Job Readiness Score.
- Develop a professional, user-friendly Flutter app interface.

0.3 Problem Statement

Job seekers in Bangladesh face challenges such as:

- Non-optimized resumes for ATS systems.
- Lack of awareness of high-demand skills.
- Difficulty in tracking skill development.
- No structured guidance to prepare for specific job roles.

CareerLens AI addresses these challenges through AI-based resume analysis, skill gap identification, personalized skill roadmap generation, and progress tracking.

0.4 Related Work

- Global Resume Analyzers: Tools like Jobscan and Rezi exist, but they are not Bangladesh-specific.
- Skill Tracking Apps: Platforms exist but lack integration with local job market trends.
- Career Guidance Platforms: Generic guidance without AI-powered resume scoring or progress tracking.

CareerLens AI improves upon these by combining resume analysis, Bangladesh-specific job insights, skill roadmap generation, and progress tracking.

0.5 Scope

CareerLens AI is aimed at Bangladeshi students and job seekers to enhance employability through AI-driven insights. The platform provides:

- Resume analysis and ATS scoring.
- Identification of missing skills for target job roles.
- Personalized skill preparation roadmap.
- Bangladesh job market insights for high-demand roles.
- Progress tracking and Job Readiness Score dashboard.

0.6 Job Market Analysis

The Bangladesh job market is growing for tech roles like Flutter Developer, Full-Stack Web Developer, AI/ML Engineer, and Data Analyst.

Job Portal	Link	Job Role / Stack	Required Skills
BD Jobs	Job1 Job2	Flutter Developer / Mobile App Developer	Flutter, Dart, Fire- base, REST API, Git, GitHub
eJobs Bangladesh	Job1 Job2	Flutter / Data Ana- lyst	Flutter, Dart, Python, SQL, Pandas, Fire- base, Git
JobMatchingBD	Job1 Job2	AI / ML Engineer	Python, Machine Learning, Tensor- Flow, Scikit-learn, Pandas, Numpy
Arc.dev & Well- found	Job1 Job2	Full-Stack Developer	React, Node.js, Fire- base, MongoDB, JavaScript, Git, REST API
Careerjet Bangladesh	Job1 Job2	Data Analyst / ML Engineer	Python, SQL, Pandas, Scikit-learn, Machine Learning, Tableau, Git

0.7 Methodology

CareerLens AI development follows Agile methodology and modular architecture.

0.7.1 Technology Stack

Layer	Technology
Frontend	Flutter (Dart)
Backend	FastAPI (Python)
ML / NLP	TF-IDF, Cosine Similarity, spaCy / NLTK
Database	Firebase / SQLite
Tools	Git, GitHub

0.7.2 Design Principles

- Mobile-First Responsive Design
- Modular architecture for scalability
- Data-driven insights from Bangladesh job market
- Transparency through visual ATS scores and progress tracking

0.7.3 Visual Models

ERD (Entity Relationship Diagram)

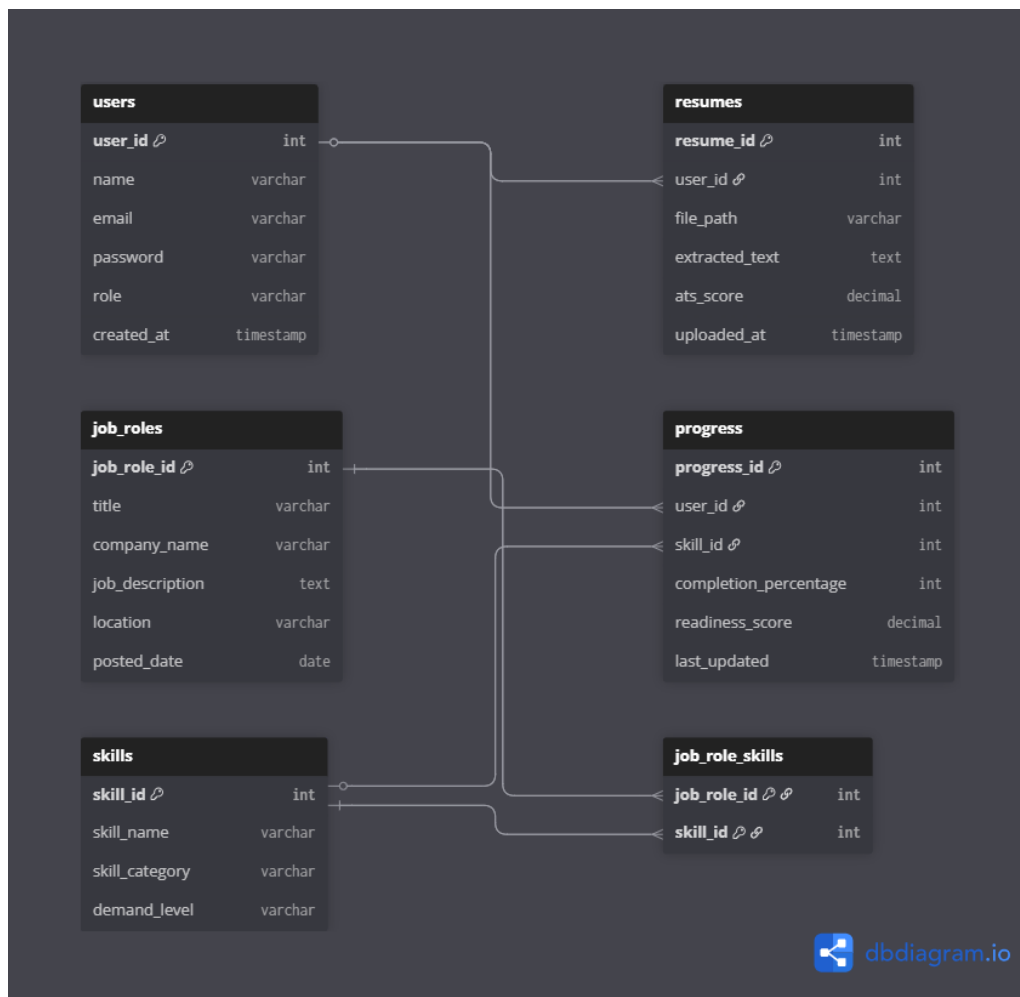


Figure 1: Entity Relationship Diagram of CareerLens AI

System Architecture Diagram

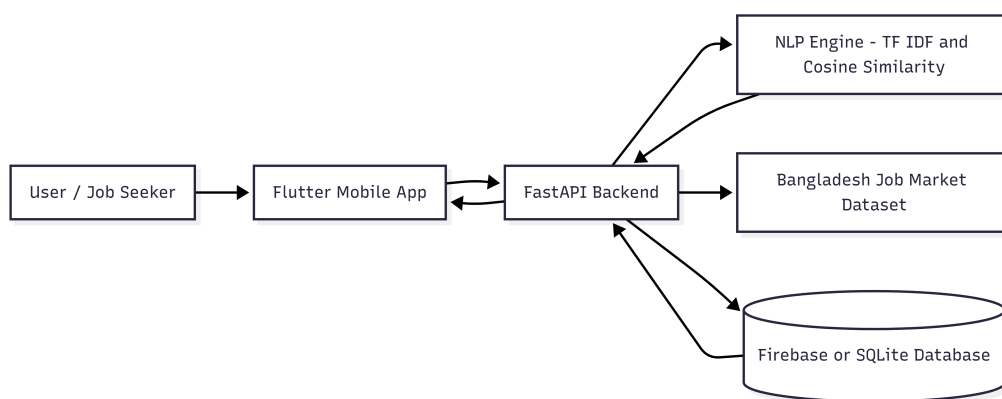


Figure 2: System architecture of CareerLens AI

0.7.4 Timeline (Gantt Chart)

Placeholder for Gantt Chart diagram.

Task	Week 1-2	Week 3-4	Week 5-6	Week 7-8
Requirement Analysis	X			
UI/UX Design		X		
Backend Setup		X	X	
ML Model Implementation			X	
Skill Roadmap			X	
Integration & Testing				X
Final Report				X

0.8 UI Mockups

Placeholder for UI mockups. - Resume Upload Screen - Job Description Input - ATS Score Dashboard - Missing Skills List - Skill Roadmap - Progress Tracker

0.9 Future Plans

- Live integration with Bangladesh job portals
- Upgrade NLP module to BERT
- Weekly skill notifications
- Employer-side dashboard
- Expand to other regional markets

0.10 Result

CareerLens AI successfully:

- Extracts skills from resumes using NLP
- Compares resumes with job descriptions using TF-IDF + Cosine Similarity
- Generates ATS scores with detailed breakdown
- Suggests missing skills and learning roadmaps
- Tracks progress and job readiness
- Provides professional Flutter-based interface

0.11 References

1. Manning, C. D., Raghavan, P., & Schütze, H. (2008). *Introduction to Information Retrieval*.
2. Jurafsky, D., & Martin, J. H. (2021). *Speech and Language Processing*, 3rd Edition.
3. BDJobs.com, eJobs.com.bd, JobMatchingBD.com, Wellfound.com, Careerjet.com.bd
4. Flutter Official Documentation: <https://flutter.dev>
5. FastAPI Documentation: <https://fastapi.tiangolo.com>

0.12 Weekly Report

No.	Week	Objective	Status	Remarks
1	1	Project proposal, research existing solutions, define scope and initial UI wireframes		
2	2	Set up Flutter environment, initialize FastAPI backend, design basic navigation		
3	3	Implement resume upload, parse resumes, integrate NLP module, connect frontend and backend		
4	4	Design database (ERD), set up Firebase/SQLite, implement data models and CRUD APIs		
5	5	Compare resumes with job descriptions, calculate ATS score, generate skill roadmap, integrate local job market		
6	6	Build dashboard, implement progress tracking and job readiness score, optimize UI		
7	7	Test all features, fix bugs, deploy app and backend, prepare final report and manual		

Table 2: Weekly development report of CareerLens AI